

Chapter 2

Limits and Their Duals

2.1 Terminal and Initial Objects

Definition 2.1.1 (Terminal Object). A *terminal object* 1 of a category $\mathcal{C}$ is an object such that for any object A , there is a unique morphism $!_A : A \longrightarrow 1$.

A category may have any number of terminal objects.

Theorem 2.1.2. A terminal object is unique up to isomorphism.

Proof. Suppose that 1 and $1'$ are terminal objects in a category $\mathcal{C}$. On one hand, since 1 is a terminal object, there is a unique morphism $!_{1'} : 1' \longrightarrow 1$. On the other hand, since $1'$ is a terminal object, there is a unique morphism $!_1 : 1 \longrightarrow 1'$.

$$1' \begin{array}{c} \xrightarrow{!_{1'}} \\ \xleftarrow{!_1} \end{array} 1$$

Then the composite $!_{1'} \circ !_1 : 1 \longrightarrow 1' \longrightarrow 1$ equals id_1 , since the morphism to 1 is unique. In the same way, the composite $!_1 \circ !_{1'} : 1' \longrightarrow 1 \longrightarrow 1'$ equals $id_{1'}$, since the morphism to $1'$ is also unique. Therefore, $!_1$ and $!_{1'}$ are mutually inverse and $1 \cong 1'$. $\square$

Definition 2.1.3 (Global Element). A *global element* of an object A in a category $\mathcal{C}$ is any morphism $1 \rightarrow A$.

Definition 2.1.4 (Initial Object). An *initial object* $\emptyset$ of a category $\mathcal{C}$ is an object such that for any object C , there is a unique morphism $!_C : \emptyset \rightarrow C$.

A category may have any number of initial objects.

Theorem 2.1.5. An initial object is unique up to isomorphism.

Proof. By the dual of the proof of Theorem 2.1.2. □

Theorem 2.1.6. For any object A and morphism $f : A \rightarrow \emptyset$, f is split epic.

Proof. Consider a morphism $f \circ !_A : \emptyset \rightarrow A \rightarrow \emptyset$. Since there is only one morphism from $\emptyset$ to any object, one has $f \circ !_A = id_\emptyset$. Thus f has a right inverse $!_A : \emptyset \rightarrow A$ and is split epic.

$$\begin{array}{ccc}
 & A & \\
 f \swarrow & & \nwarrow !_A \\
 \emptyset & \xleftarrow{id_\emptyset} & \emptyset
 \end{array}$$

□

Exercise 2.1.7. Suppose that there is a terminal object 1 . Show that every arrow $f : 1 \rightarrow B$ is monic.

Exercise 2.1.8. An object which is both terminal and initial is called a *zero object*. Show that, if there is an initial object $\emptyset$, a terminal object 1 , and an arrow $1 \rightarrow \emptyset$, then 1 and $\emptyset$ are zero objects.

2.2 Products and Coproducts

Definition 2.2.1 (Binary Product). A *binary product* of objects A and B in $\mathcal{C}$ is specified by:

- an object $A \times B$ of $\mathcal{C}$
- two projections: morphisms $\pi_1 : A \times B \rightarrow A$ and $\pi_2 : A \times B \rightarrow B$, such that given any object C and morphisms $f : C \rightarrow A$, $g : C \rightarrow B$, there is a unique morphism $\langle f, g \rangle : C \rightarrow A \times B$ for which $\pi_1 \circ \langle f, g \rangle = f$ and $\pi_2 \circ \langle f, g \rangle = g$, i.e. the following diagram commutes.

$$\begin{array}{ccccc}
 & & C & & \\
 & f \swarrow & \downarrow \langle f, g \rangle & \searrow g & \\
 A & \xleftarrow{\pi_1} & A \times B & \xrightarrow{\pi_2} & B
 \end{array}$$

We shall simply refer to a binary product $A \times B$ instead of the triple $\langle A \times B, \pi_1, \pi_2 \rangle$.

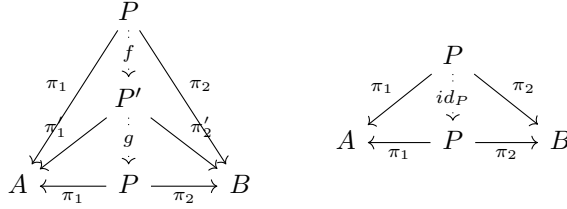
Exercise 2.2.2. Show that $\langle f, g \rangle$ is monic if f is monic.

Theorem 2.2.3. For any objects A and B , a binary product of A and B is unique up to isomorphism if it exists.

Proof. Suppose that P and P' are binary products of A and B . By Definition 2.2.1, there is a unique arrow $f : P \rightarrow P'$ such that $\pi'_1 \circ f = \pi_1$ and $\pi'_2 \circ f = \pi_2$, and there is a unique arrow $g : P' \rightarrow P$ such that $\pi_1 \circ g = \pi'_1$ and $\pi_2 \circ g = \pi'_2$.

$$\begin{array}{ccccc}
 & & P & & \\
 & \pi_1 \swarrow & \downarrow f & \searrow \pi_2 & \\
 A & \xleftarrow{g} & P' & \xrightarrow{f} & B \\
 & \nwarrow \pi'_1 & \uparrow \pi'_2 & & \\
 & & P & &
 \end{array}$$

Therefore, the composite $g \circ f$ is a morphism from P to P that makes the left diagram below commute, namely, $\pi_1 \circ (g \circ f) = (\pi_1 \circ g) \circ f = \pi'_1 \circ f = \pi_1$ and $\pi_2 \circ (g \circ f) = (\pi_2 \circ g) \circ f = \pi'_2 \circ f = \pi_2$.



On the other hand, the right diagram above commutes as well. Since P is a binary product of A and B , $g \circ f = id_P$. Similarly, $f \circ g = id_{P'}$. Thus, g and f are mutually inverse, and $P \cong P'$. $\square$

Theorem 2.2.4. For any objects A, B , $A \times B \cong B \times A$.

Proof. Both $\langle A \times B, \pi_1, \pi_2 \rangle$ and $\langle B \times A, \pi_2, \pi_1 \rangle$ are binary products of A and B . Therefore, by Theorem 2.2.3, one has $A \times B \cong B \times A$. $\square$

Exercise 2.2.5. Show that the product constructions are associative: for any objects A, B, C , $A \times (B \times C) \cong (A \times B) \times C$

Definition 2.2.6 (Product of Morphisms). Let $\mathcal{C}$ be a category with finite products and take morphisms $f : A \rightarrow B$ and $f' : A' \rightarrow B'$. We write $f \times f' : A \times A' \rightarrow B \times B'$ for the morphism $\langle f\pi, f'\pi' \rangle$ where $\pi : A \times A' \rightarrow A$ and $\pi' : A \times A' \rightarrow A'$.

$$\begin{array}{ccccc}
 A & \xleftarrow{\pi} & A \times A' & \xrightarrow{\pi'} & A' \\
 f \downarrow & & f \times f' \downarrow & & \downarrow f' \\
 B & \xleftarrow{\pi} & B \times B' & \xrightarrow{\pi'} & B'
 \end{array}$$

Exercise 2.2.7. Show that each of the following equations hold ($f : A \longrightarrow B$, $f' : A \longrightarrow B'$, $g : B \longrightarrow C$, and $g' : B' \longrightarrow C'$).

$$\begin{aligned} id_A \times id_{A'} &= id_{A \times A'} \\ (g \times g') \circ (f \times f') &= g \circ f \times g' \circ f' \\ (g \times g') \circ \langle f, f' \rangle &= \langle g \circ f, g' \circ f' \rangle \end{aligned}$$

Theorem 2.2.8. In a category $\mathcal{C}$ with binary products, any object A is isomorphic to $1 \times A$ ($A \cong 1 \times A$).

Proof. Consider any morphism $f : A \longrightarrow A$ such that $id_A = id_A \circ f$, which means $f = id_A$. Thus, id_A is the unique morphism that makes the following diagram commute.

$$\begin{array}{ccc} & A & \\ !_A \swarrow & \downarrow id_A & \searrow id_A \\ 1 & \xleftarrow{!_A} A & \xrightarrow{id_A} A \end{array}$$

Therefore, $\langle A, !_A, id_A \rangle$ is a binary product of 1 and A . By Theorem 2.2.3, $A \cong 1 \times A$. $\square$

Definition 2.2.9 (Graph). Suppose that objects A and B have a product $A \times B$. Take any arrow $f : A \longrightarrow B$, the *graph of f* is defined as the arrow Γ_f that makes the following diagram commute:

$$\begin{array}{ccccc} & & A & & \\ & id_A \swarrow & \downarrow \Gamma_f & \searrow f & \\ A & \xleftarrow{\pi_1} & A \times B & \xrightarrow{\pi_2} & B \end{array}$$

Definition 2.2.10 (Diagonal). For any object A , the *diagonal* $\Delta_A : A \longrightarrow A \times A$ is a graph of id_A .

$$\begin{array}{ccccc}
 & & A & & \\
 & \swarrow id_A & \downarrow \Delta_A & \searrow id_A & \\
 A & \xleftarrow{\pi_1} & A \times A & \xrightarrow{\pi_2} & A
 \end{array}$$

Exercise 2.2.11. Show that, if a category $\mathcal{C}$ has binary products, $\langle h, k \rangle = (h \times k) \circ \Delta_T$ for any $h : T \longrightarrow A$ and $k : T \longrightarrow B$.

Definition 2.2.12 (Binary Coproduct). A *binary coproduct* of objects A and B in $\mathcal{C}$ is specified by:

- an object $A + B$ of $\mathcal{C}$
- two projections: morphisms $\iota_1 : A \longrightarrow A + B$ and $\iota_2 : B \longrightarrow A + B$, such that given any object C and morphisms $f : A \longrightarrow C$, $g : B \longrightarrow C$, there is a unique morphism $\begin{pmatrix} f \\ g \end{pmatrix} : A + B \longrightarrow C$ for which $\begin{pmatrix} f \\ g \end{pmatrix} \circ \iota_1 = f$ and $\begin{pmatrix} f \\ g \end{pmatrix} \circ \iota_2 = g$, i.e. the following diagram commutes.

$$\begin{array}{ccccc}
 A & \xrightarrow{\iota_1} & A + B & \xleftarrow{\iota_2} & B \\
 & \searrow f & \downarrow \begin{pmatrix} f \\ g \end{pmatrix} & \swarrow g & \\
 & & C & &
 \end{array}$$

Theorem 2.2.13. For any objects A and B , a binary coproduct of A and B is unique up to isomorphism if it exists.

Theorem 2.2.14. In a category $\mathcal{C}$ with binary coproducts, any object A is isomorphic to $\emptyset + A$ ($A \cong \emptyset + A$).

Theorem 2.2.15. For any objects A, B , $A + B \cong B + A$.

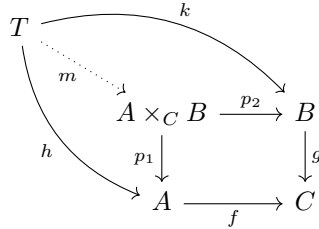
Proof. Reverse the arrows in the proofs of Theorem 2.2.3, Theorem 2.2.8, and Theorem 2.2.4, respectively. $\square$

Exercise 2.2.16. Show that the coproduct constructions are associative: for any objects A, B, C , $A + (B + C) \cong (A + B) + C$

2.3 Pullbacks and Pushouts

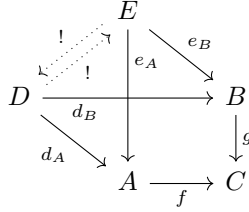
Definition 2.3.1 (Pullback). A *pullback* for a pair of morphisms $f : A \rightarrow C$ and $g : B \rightarrow C$ in a category $\mathcal{C}$ is given by:

- an object $A \times_C B$ of $\mathcal{C}$
- a pair of morphisms $p_1 : A \times_C B \rightarrow A$ and $p_2 : A \times_C B \rightarrow B$ (called *projection arrows*) for which $f \circ p_1 = g \circ p_2$ such that given any two morphisms $h : T \rightarrow A$ and $k : T \rightarrow B$ in $\mathcal{C}$ for which $f \circ h = g \circ k$, there is a unique morphism $m : T \rightarrow A \times_C B$ for which $p_1 \circ m = h$ and $p_2 \circ m = k$, i.e. the following diagram commutes.

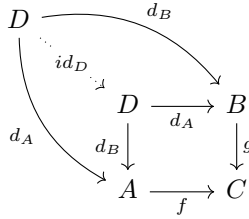


Theorem 2.3.2. A pullback for a given pair of morphisms is determined up to isomorphism.

Proof. Suppose that $D, d_A : D \rightarrow A, d_B : D \rightarrow B$ and $E, e_A : E \rightarrow A, e_B : E \rightarrow B$ are pullbacks for a pair of morphisms $f : A \rightarrow C$ and $g : B \rightarrow C$. By the definition of pullback, there is a unique arrow from D to E and vice versa.



On the other hand, since D is a pullback, the arrow from D to D that makes the following diagram commute is unique, that is, id_D .



This means that the arrow $D \xrightarrow{!} E \xrightarrow{!} D$ is id_D . Similarly, the arrow $E \xrightarrow{!} D \xrightarrow{!} E$ is id_E . Hence $D \cong E$. $\square$

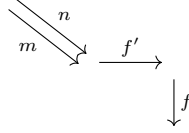
Exercise 2.3.3. Show that, for any arrow $f : A \rightarrow B$ and object C , if A and B have a product with C , then the following square is a pullback.

$$\begin{array}{ccc}
 A \times C & \xrightarrow{f \times id_C} & B \times C \\
 \pi_1 \downarrow & & \downarrow \pi_1 \\
 A & \xrightarrow{f} & B
 \end{array}$$

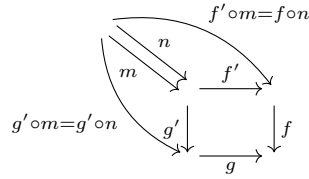
Theorem 2.3.4. If the following diagram is a pullback and g is monic, then f' is also monic.

$$\begin{array}{ccc}
 & \xrightarrow{f'} & \\
 g' \downarrow & & \downarrow f \\
 & \xrightarrow{g} &
 \end{array}$$

Proof. For any two arrows m, n such that $f' \circ m = f' \circ n$, we have $f \circ f' \circ m = f \circ f' \circ n$.

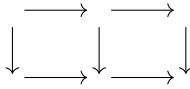


On the other hand, $f \circ f' = g \circ g'$. So we also have $g \circ g' \circ m = g \circ g' \circ n$. This means that the following diagram commutes.

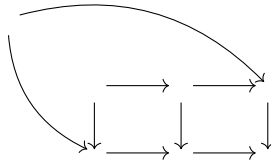


Since the square is a pullback, $m = n$. □

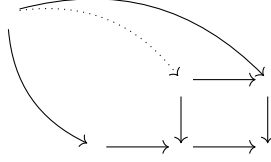
Theorem 2.3.5. In the following diagram, the outer rectangle is a pullback if the two inner squares are pullbacks.



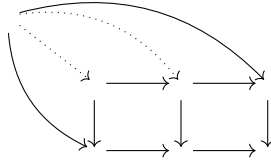
Proof. Consider any two arrows that make the following diagram commute.



Since the right square is a pullback, there is a unique arrow to the upper left corner of it.



Then, since the left square is also a pullback, there is a unique arrow to the upper left corner of it. Therefore, the whole square is a pullback.



□

Exercise 2.3.6. Suppose that there is a commutative triangle

$$\begin{array}{ccc} C & \xrightarrow{h} & D \\ & \searrow m & \swarrow n \\ & B & \end{array}$$

and suppose that m and n have the following pullbacks along f :

$$\begin{array}{ccc} E & \xrightarrow{q} & C \\ p \downarrow & & \downarrow m \\ A & \xrightarrow{f} & B \end{array} \quad \begin{array}{ccc} F & \xrightarrow{s} & D \\ r \downarrow & & \downarrow n \\ A & \xrightarrow{f} & B \end{array}$$

Show that, then, there is a unique $u : E \rightarrow F$ that makes the following diagram commute, and E, u, q a pullback for h and s :

$$\begin{array}{ccccc} E & \xrightarrow{q} & C & & \\ & \searrow u & & \searrow h & \\ & & F & \xrightarrow{s} & D \\ p \searrow & & \swarrow r & & \swarrow m \\ & A & \xrightarrow{f} & B & \\ & & & \swarrow n & \end{array}$$

2.4 Equalizers and Coequalizers

Definition 2.4.1 (Equalizer). A morphism $e : E \longrightarrow A$ is an *equalizer* for a pair of morphisms $f, g : A \longrightarrow B$ in a category $\mathcal{C}$ iff

- $f \circ e = g \circ e$
- given any $h : T \longrightarrow A$ with $f \circ h = g \circ h$, there is a unique $k : T \longrightarrow E$ for which $h = e \circ k$.

i.e. the following diagram commutes.

$$\begin{array}{ccccc} E & \xrightarrow{e} & A & \xrightleftharpoons[g]{f} & B \\ \uparrow k & \nearrow h & & & \\ T & & & & \end{array}$$

Theorem 2.4.2. Every equalizer is monic.

Proof. Suppose that e is an equalizer of f and g in the following diagram, i.e. $f \circ e = g \circ e$.

$$E \xrightarrow{e} A \xrightleftharpoons[g]{f} B$$

Consider two morphisms h and k such that $e \circ h = e \circ k$.

$$\begin{array}{ccc} E & \xrightarrow{e} & A \\ \uparrow h \uparrow k & \nearrow e \circ h = e \circ k & \\ T & & \end{array}$$

Since $f \circ e = g \circ e$, it follows that $f \circ e \circ h = g \circ e \circ h$. Then there is a unique arrow from T to E that makes the following diagram commute.

$$\begin{array}{ccc} E & \xrightarrow{e} & A \xrightleftharpoons[g]{f} B \\ \uparrow & \nearrow e \circ h & \\ T & & \end{array}$$

Since $e \circ h = e \circ k$, it turns out that $h = k$. Therefore e is monic. $\square$

Theorem 2.4.3. If a category has all binary products and equalizers for any two morphisms, then it has a pullback for any two morphisms $f : A \rightarrow C$ and $g : B \rightarrow C$.

Exercise 2.4.4. Prove Theorem 2.4.3 by showing that the equalizer for $f \circ \pi_1$ and $g \circ \pi_2$ gives a pullback diagram for $f : A \rightarrow C$ and $g : B \rightarrow C$.

$$\begin{array}{ccccc}
 T & & & & B \\
 & \searrow e & & \nearrow \pi_2 & \downarrow g \\
 & & A \times B & & \\
 & \nearrow \pi_1 & & & \\
 A & \xrightarrow{f} & & & C
 \end{array}$$

Exercise 2.4.5. Suppose that a category has a pullback for every pair $f : A \rightarrow C$ and $g : B \rightarrow C$ and all binary products. Then it has an arrow from $A \times_C B$ to $A \times B$ and it is an equalizer for $f \circ \pi_1$ and $g \circ \pi_2$.

Theorem 2.4.6. The graph of $f : A \rightarrow B$ is an equalizer for $f \circ \pi_1 : A \times B \rightarrow B$ and $\pi_2 : A \times B \rightarrow B$.

Exercise 2.4.7. Prove Theorem 2.4.6.

2.5 Kernel and Cokernel Pairs

Definition 2.5.1 (Kernel Pair). For any morphism $f : A \rightarrow B$, a *kernel pair* for f is the pair of projection arrows in a pullback of f along itself.

$$\begin{array}{ccc}
 K & \xrightarrow{h} & A \\
 k \downarrow & & \downarrow f \\
 A & \xrightarrow{f} & B
 \end{array}$$

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ f \downarrow & & \downarrow h \\ B & \xrightarrow{k} & C \end{array}$$

Theorem 2.5.3. Kernel pairs for any morphism are determined up to isomorphism.

☐



2.6 Cones and Cocones

Definition 2.6.1 (Cone). (McLarty, 1992, p.48) Consider a diagram D with an object A_i at each vertex i , and an arrow f_e at each edge e . A *cone over D* consists of an object C and morphisms $p_i : C \rightarrow A_i$ to each vertex such that, for every edge e with morphism $f_e : A_i \rightarrow A_j$ we have $f_e \circ p_i = p_j$.

Definition 2.6.2 (Limit). A *limit* for the diagram D is a cone over D (an object C and morphisms p_i) which satisfy the following properties:

- For any cone (say, T and q_i) over D , there is a unique arrow $u : T \rightarrow C$, such that $p_i \circ u = q_i$ for each vertex i .

The morphisms p_i are called *projection arrows*.

Example 2.6.3. Consider a diagram which has no vertices. Then a cone over it is a limit iff it is a terminal object.

$$T \xrightarrow{\quad u \quad} C$$

Example 2.6.4. Consider a diagram which has two vertices and no edges.

$$A_1 \qquad A_2$$

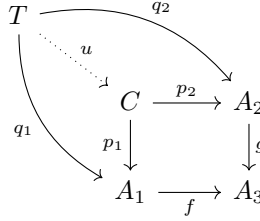
Then a cone over it is a limit iff it is a product diagram.

$$\begin{array}{ccccc} & & T & & \\ & q_1 \swarrow & \vdots & \searrow q_2 & \\ & & u & & \\ & & \downarrow & & \\ A_1 & \xleftarrow{p_1} & C & \xrightarrow{p_2} & A_2 \end{array}$$

Example 2.6.5. Consider a diagram which has three vertices and two edges.

$$\begin{array}{ccc} & A_2 & \\ & \downarrow g & \\ A_1 & \xrightarrow{f} & A_3 \end{array}$$

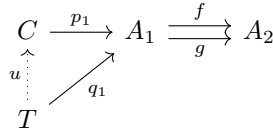
Then a cone over it is a limit iff it is a pullback diagram.



Example 2.6.6. Consider a diagram which has two vertices and two edges between them.

$$A \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} B$$

Then a cone over it is a limit iff it is an equalizer diagram.



Cocone is the dual notion of a cone, and *colimits* are the dual notion of limits.

Definition 2.6.7 (Cocone). (McLarty, 1992, p.50) A *cocone* on the diagram D is an object C together with arrows $q_i : A_i \rightarrow C$ for each vertex i , such that for every edge e with arrow $f_e : A_i \rightarrow A_j$ we have $q_j \circ f_e = q_i$.

Definition 2.6.8 (Colimit). (McLarty, 1992, p.50) A *colimit* for the diagram D is a cocone on D , namely C with arrows p_i , with the following property: given any cocone T , $r_i : A_i \rightarrow T$ on D , there is exactly one arrow $u : C \rightarrow T$ such that for every vertex i , $u \circ p_i = r_i$.

2.7 All Finite Limits and Colimits

Definition 2.7.1 (Finite Products). A category $\mathcal{C}$ has *finite* products iff it has products for every finite diagram.

Definition 2.7.2 (All Finite Limits). A category $\mathcal{C}$ has *all finite limits* iff $\mathcal{C}$ has a limit for every finite diagram.

Theorem 2.7.3. If a category has a terminal object, binary products, and an equalizer for every parallel pair of arrows, then it has all finite limits.

Proof. Suppose that there is a product $\Pi_i A_i$ of all the A_i , with projection $p_j : \Pi_i A_i \rightarrow A_j$ for each vertex j . Use the notation A_e for the codomain of the arrow at edge e , so that if $f_e : A_j \rightarrow A_k$ then $A_e = A_k$. Suppose that there is a product $\Pi_e A_e$ with one factor A_e for each edge e , and a projection $p_e : \Pi_e A_e \rightarrow A_k$ for each $f_e : A_j \rightarrow A_k$.

Let $r : \Pi_i A_i \rightarrow \Pi_e A_e$ be the arrow such that, for each edge e with $f_e : A_j \rightarrow A_k$, the following triangle commutes:

$$\begin{array}{ccc} \Pi_i A_i & \xrightarrow{r} & \Pi_e A_e \\ & \searrow p_k & \downarrow p_e \\ & & A_k \end{array}$$

Let $s : \Pi_i A_i \rightarrow \Pi_e A_e$ be the arrow such that for each edge e the following square commutes:

$$\begin{array}{ccc} \Pi_i A_i & \xrightarrow{s} & \Pi_e A_e \\ p_j \downarrow & & \downarrow p_e \\ A_j & \xrightarrow{f_e} & A_k \end{array}$$

Then the diagram D has a limit iff there is an equalizer for r and s . More specifically, $l : L \rightarrow \Pi_i A_i$ is an equalizer for r and s iff L together with all $p_i \circ l : L \rightarrow A_i$ forms a limit cone for D .

Consider any cone $T, h_i : T \rightarrow A_i$ over D . There is a unique $h : T \rightarrow \Pi_i A_i$ such that for each vertex k we have $p_k \circ h = h_k$. Then $r \circ h = s \circ h$ iff for each edge

$e, p_e \circ r \circ h = p_e \circ s \circ h$. But, by the definition of r, s and h , that last equation means that $h_k = f_e \circ h_j$, where $f_e : A_j \rightarrow A_k$. Thus T, h factors uniquely through L, l iff the cone T, h_i factors uniquely through the cone $L, p_i \circ l$. $\square$

Theorem 2.7.4. If a category has a terminal object and pullbacks for all corner of arrows, then it has all finite limits.

Proof. By Theorem 2.7.3, Theorem 2.7.5 and Theorem 2.7.7. $\square$

Theorem 2.7.5. A category with a terminal object and pullbacks for all corners of arrows, has a product for each pair of objects.

Exercise 2.7.6. Prove Theorem 2.7.5 by showing that $\langle A \times_1 B, p_1, p_2 \rangle$ in the following diagram is also a product diagram for A and B .

$$\begin{array}{ccc} A \times_1 B & \xrightarrow{p_2} & B \\ p_1 \downarrow & & \downarrow !_B \\ A & \xrightarrow{!_A} & 1 \end{array}$$

Theorem 2.7.7. A category with a product for each pair of objects, and a pullback for every corner of arrows, has an equalizer for every parallel pair.

Exercise 2.7.8. Prove Theorem 2.7.7 by showing that the following e is an equalizer for any pair $f, g : A \rightarrow B$.

$$\begin{array}{ccc} E & \xrightarrow{p_2} & B \\ e \downarrow & & \downarrow \Delta_B \\ A & \xrightarrow{\langle f, g \rangle} & B \times B \end{array}$$

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